

[Date]

Assignment-1

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1. List three real-life scenarios where a multimeter would be used to troubleshoot a computer or smartphone issue, and explain in one line how the multimeter helps in each case.

❖ **Three Real-Life Uses of a Multimeter**

1. **Computer Not Turning On:** A multimeter checks the power supply voltage to identify power-related problems.
2. **Phone/Laptop Not Charging:** It checks the charging port and adapter voltage to find charging faults.
3. **Device Shutting Down Frequently:** It checks the battery voltage and power circuit to identify power issues.

Multimeter: A multimeter is mainly used to measure voltage, current, resistance, and continuity in electrical and electronic circuits.

2. Watch a short YouTube video demo of a POST card in action (search for 'POST card motherboard test'), then write a brief summary (3-4 lines) describing what the POST card displays and what problem it helped identify in the video. Focus on the codes shown and how they relate to hardware issues.

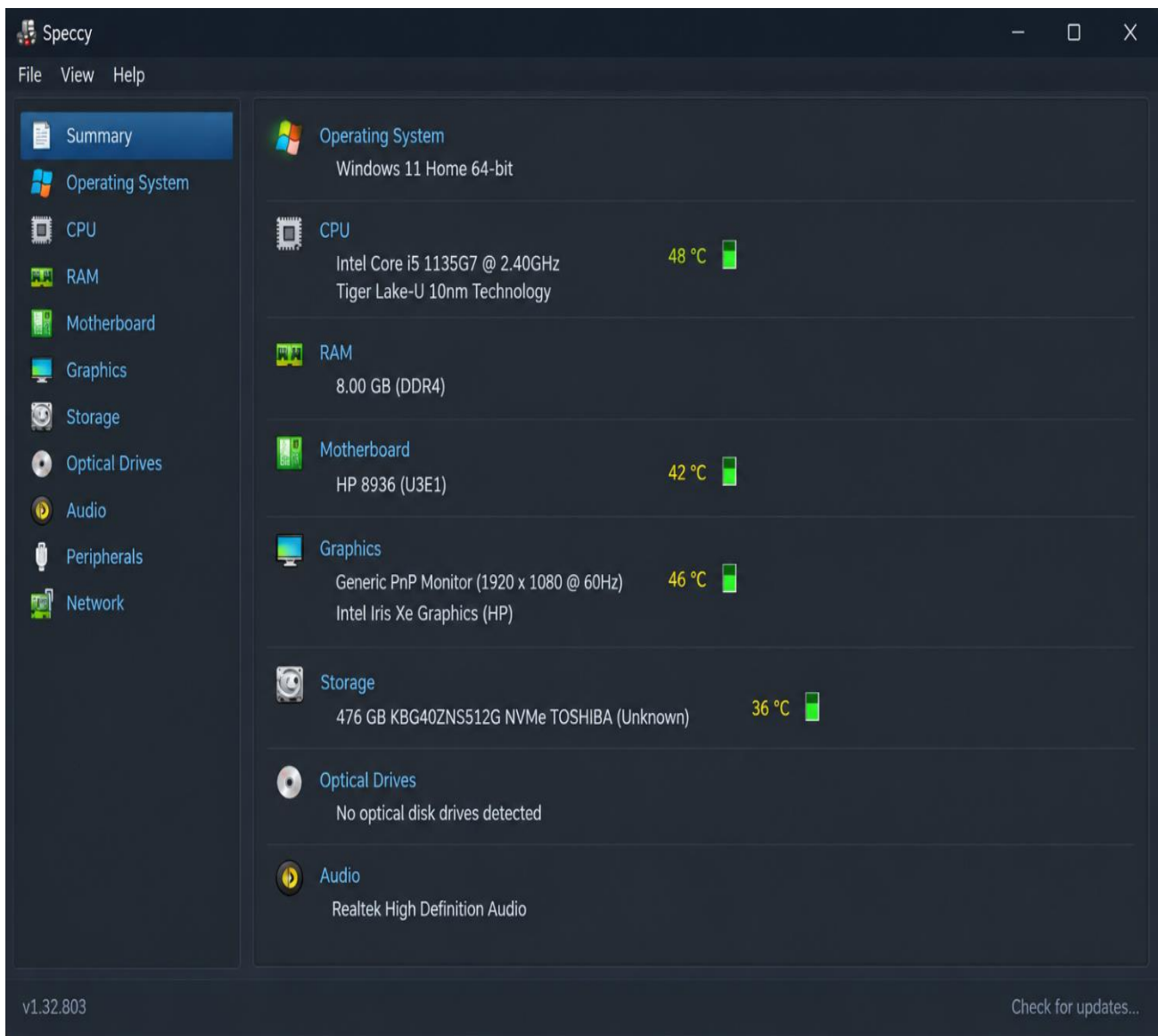
- ❖ The POST card displays hexadecimal codes during the computer's startup (POST) process.

In the demo, codes such as 3387, 2900, and AA00 were displayed as the BIOS performed its checks.

The codes showed that the BIOS check and boot process were completing successfully.

POST cards help identify hardware problems by showing the last code when the startup process stops

3. Download and install any free software diagnostic tool (such as HWMonitor, Speccy, or CrystalDiskInfo), run it on your laptop or desktop, and take a screenshot of the main dashboard showing system health details.



Tool	What it checks	Example Problem Detected
Multimeter	Checks electrical values like voltage, resistance, and continuity.	Dead power supply or broken cable

POST Card	Checks the computer's startup/POST process and shows error codes.	Faulty RAM or motherboard problem
Software Diagnostics	Checks hardware health, temperatures, storage, RAM, CPU, etc.	Overheating CPU or failing hard drive

4. Compare the use of a multimeter, POST card, and software diagnostics by creating a table with three columns (Tool, What it checks, Example Problem Detected) and fill in at least one example for each tool.